1 (a) Write $x^2 - 6x + 1$ in the form $(x + a)^2 + b$ where a and b are integers. (2)

(b) Hence, or otherwise, write down the coordinates of the turning point of the graph of $y = x^2 - 6x + 1$

(1)
(3 marks)

2 (a) Write $x^2 + 8x + 5$ in the form $(x + a)^2 + b$ where a and b are integers.

(2)

(b) Hence, or otherwise, write down the coordinates of the turning point of the graph of $y = x^2 + 8x + 5$

(1)
(3 marks)

3 (a) Write $x^2 + 10x + 2$ in the form $(x + a)^2 + b$ where a and b are integers. (2)

(b) Hence, or otherwise, write down the coordinates of the turning point of the graph of $y = x^2 + 10x + 2$

(1)
(3 marks)

4 (a) Write $x^2 - 2x - 1$ in the form $(x + a)^2 + b$ where a and b are integers.

(2)

(b) Hence, or otherwise, write down the coordinates of the turning point of the graph of $y = x^2 - 2x - 1$

(1)
(3 marks)

5 By completing the square, find the coordinates of the turning point of the curve with the equation $y = x^2 + 8x + 3$

You must show all your working.

(3 marks)

6 By completing the square, find the coordinates of the turning point of the curve with the equation $y = x^2 + 10x - 8$

You must show all your working.

(3 marks)

7 By completing the square, find the coordinates of the turning point of the curve with the equation $y = x^2 + 3x - 7$

You must show all your working.

(3 marks)

8 By completing the square, find the coordinates of the turning point of the curve with the equation $y = x^2 - x + 8$

You must show all your working.

(3 marks)

9 (a) Write $2x^2 - 12x + 23$ in the form $a(x + b)^2 + c$ where a, b, and c are integers.

(3)

(b) Hence, or otherwise, write down the coordinates of the turning point of the graph of $y = 2x^2 - 12x + 23$

(1)
(4 marks)

10 (a) Write $2x^2 + 16x + 26$ in the form $a(x + b)^2 + c$ where a, b, and c are integers.

(3)

(b) Hence, or otherwise, write down the coordinates of the turning point of the graph of $y = 2x^2 + 16x + 26$

(1)

(4 marks)

11 (a) Write $3x^2 - 6x + 6$ in the form $a(x + b)^2 + c$ where a, b, and c are integers.

(3)

(b) Hence, or otherwise, write down the coordinates of the turning point of the graph of $y = 3x^2 - 6x + 6$

(1)

(4 marks)

12 (a) Write $3x^2 - 30x + 63$ in the form $a(x + b)^2 + c$ where a, b, and c are integers.

(3)

(b) Hence, or otherwise, write down the coordinates of the turning point of the graph of $y = 3x^2 - 30x + 63$

(1)

(4 marks)

13 By completing the square, solve $x^2 + 10x - 3 = 0$ Give your answers in surd form.

(5 marks)

14 By completing the square, solve $x^2 + 5x + 4.25 = 0$ Give your answers in surd form.

(5 marks)